

Raghwendra Mahato

Kathmandu, Nepal | +977 9848867593 | raghwendramahato@gmail.com
raghwendramahato.com.np | <https://github.com/Raghav101010>

Professional Summary

AI and Machine Learning Engineer with a background in Electronics and Communication Engineering and over 12 years of experience in telecom project management. Skilled in building end-to-end AI applications including deep learning model development, REST API deployment, and full-stack AI systems using FastAPI, React, and Streamlit.

Technical Skills

Programming: Python, SQL

Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, Support Vector Machine, Naive Bayes, Time Series, RNN, LSTM, GRU

Deep Learning: CNN, Transfer Learning, Computer Vision, Transformers, Encoder Decoder

Frameworks: TensorFlow, Scikit-learn, Pandas, NumPy, Pytorch

Computer Vision: YOLOv8, DeepSORT, Object Detection, Multi-Object Tracking, OpenCV

Backend: FastAPI, REST APIs

Frontend: React, Streamlit

Deployment: Docker, Render, Railway, Vercel, Streamlit Cloud, Hugging Face Spaces

Tools: Git, GitHub, Jupyter Notebook, Google Colab

Professional Experience

Head of IT Department — International Business Network, Bhaisepati, Lalitpur

April 2023 – Present

Telecom Project Management (12 Years)

Managed telecom infrastructure and network deployment projects, coordinating cross-functional teams, vendors, and stakeholders. Developed strong analytical, problem-solving, and project delivery skills.

Data Science Engineer — Sirpi Technology, Bangalore

July 2022 – March 2023

- Developed machine learning models for predictive analytics.
- Built preprocessing pipelines and feature engineering workflows.
- Implemented REST APIs using FastAPI to deploy models.
- Collaborated with teams to integrate AI solutions into applications.

AI / Machine Learning Projects

1. Real-Time Face Mask Detection System

Developed a real-time computer vision system that detects whether individuals are wearing face masks using a convolutional neural network and live webcam input.

- Performs face detection and mask classification in real time using a trained CNN model.
- Interactive analytics dashboard built with Streamlit to visualize detection counts.
- Configured with dual remotes: GitHub (origin) and Hugging Face (hf) for automated deployment on Hugging Face Spaces.
- GitHub repository structured with main and raghav branches supporting different deployment environments.
- Full application including 121 MB trained model containerized with Docker and deployed on Railway.
- Due to GitHub file size limitations, the trained model is downloaded at runtime from Google Drive.
- WebRTC browser restrictions affect webcam streaming in some cloud environments, though the system runs fully when executed locally.
- Analytics counter reflects frame-based detections rather than unique individuals; future improvements include implementing tracking algorithms such as DeepSort or ByteTrack.

2. Potato Leaf Disease Detection System

Built an end-to-end deep learning web application that classifies potato leaf diseases from uploaded images using convolutional neural networks.

- Trained a CNN model using TensorFlow to classify Early Blight, Late Blight, and Healthy leaves.
- Designed a high-performance backend API using FastAPI for real-time model inference.
- Developed an intuitive frontend interface using React for image upload and prediction visualization.
- Frontend deployed on Vercel for fast global delivery.
- Backend API and model inference deployed on Render.
- Application follows REST API architecture where the frontend communicates with backend services using HTTP requests.
- Image preprocessing, model inference, and prediction generation are handled within the API pipeline.

3. People Detection, Tracking and Bidirectional Counting System

Designed a real-time video analytics system that detects and tracks people using modern computer vision techniques.

- Used YOLOv8 for accurate person detection and DeepSORT for multi-object tracking.
- Assigned unique IDs to individuals and tracked movement across frames.

- Implemented line-based bi-directional counting to determine UP or DOWN movement.
- Designed modular architecture separating detection, tracking, counting, and video processing components.
- Applied performance optimizations including frame skipping and frame resizing to improve inference speed.
- System monitors inference FPS and display FPS for real-time performance insights.
- Annotated output video includes bounding boxes, tracking IDs, and directional counters.
- Interactive Streamlit web interface allows users to upload videos and view processed results.
- Application deployed on Streamlit Community Cloud for browser-based access.

4. Multimodal Image Captioning and Visual Question Answering System (Ongoing)

- Developing a multimodal AI system generating captions for images.
- Integrates vision encoders with language models for caption generation.
- Supports visual question answering about image content.

Education

Bachelor's Degree in Electronics and Communication Engineering